

CLEANING DATA: ITS 80% OF WHAT DATA SCIENTISTS DO!

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DATA SCIENTIST



What my friends think I do



What my mom thinks I do



What society thinks I do



What my company thinks I do



What I think I do



What I actually do

First Principles in Data Cleaning

- NEVER use Excel
- ALWAYS use code
 - Reproducible
 - Can find 'mistakes'
 - Easy to go back and fix data
- Always keep raw original data
 - This is your 'hard copy' original
- Updated data save as separate file in separate folder
- Date time stamp updated & cleaned data

Biggest Tasks in Cleaning Data

- Get to know your data (i.e., explore)
- Evaluate biological plausible ranges
- Make sure case sensitivity is correct
- Make sure the correct rows/levels are integrated
- Make sure strings are coherent

Get to know your data (i.e., explore)

- Talk to the data collectors
 - Understand the data context
 - Any peculiarities with the data
- Have a pre-meeting with data leaders
 - Any questions on formatting
 - Data dictionary questions
- Visualize and create shell tables of basic data

Good Practice First Steps

- View the data
 - Overall
 - 'Head'
 - 'Tail'
- Get the names of each variable
- Look at the structure of each variable

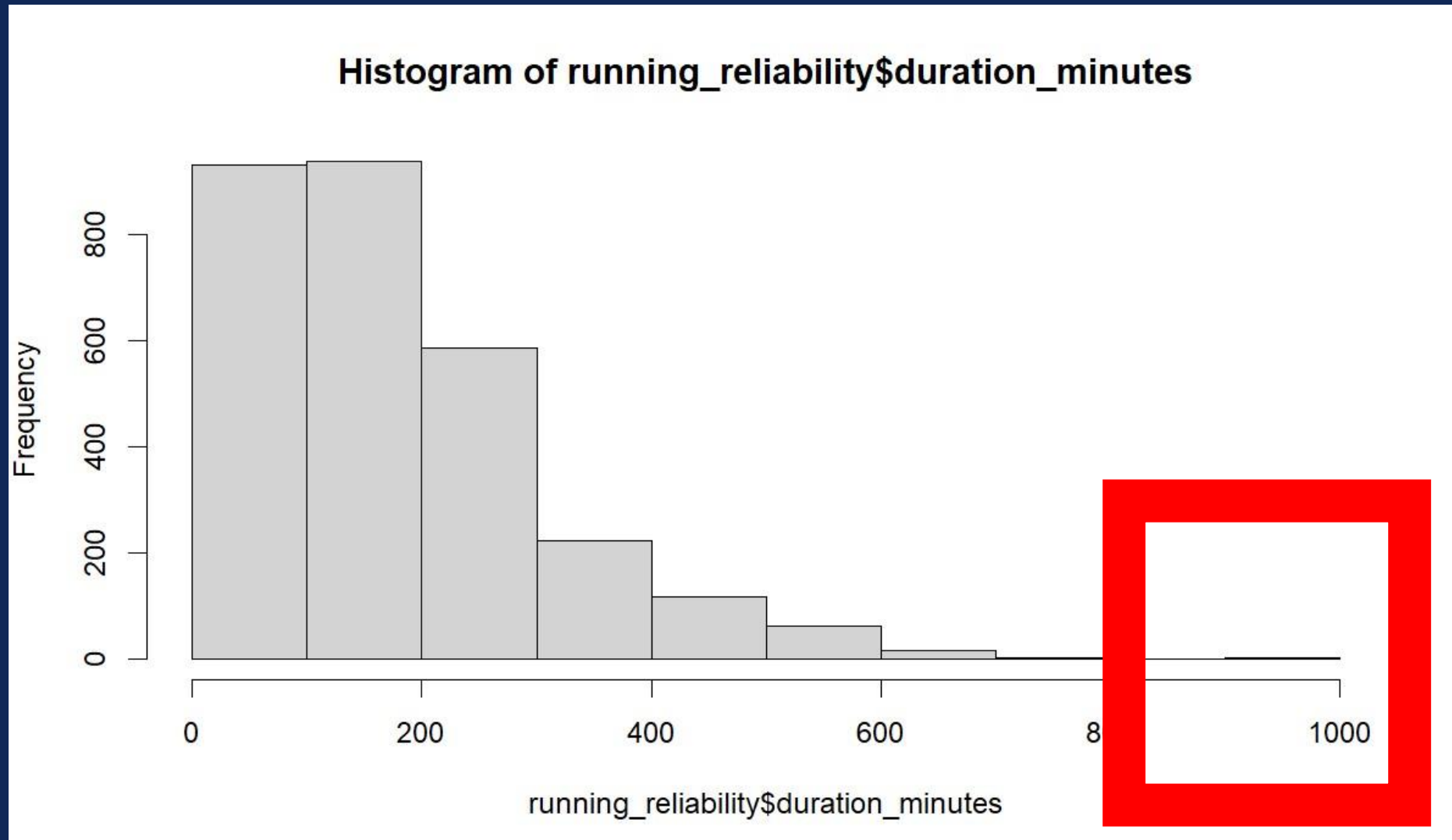
Good Practice First Steps

- View the data
 - `View(dataframe)`
- Head
 - `head(dataframe, n = 10)`
- Tail
 - `tail(dataframe, n = 10)`
- Names
 - `names(dataframe)`
- Structure
 - `str(dataframe)`

Evaluate biological plausible ranges

- Visualize through histograms & kernel density plots
- Filter extreme values for plausibility
- Check against other variables that are related

```
hist(running_reliability$duration_minutes)
```



```
running_reliability %>%  
filter(duration_minutes > 800) %>%  
select(redcap_record_id, week, distance_kilometers, duration_minutes)
```

redcap_record_id	week	distance_kilometers	duration_minutes
1341	14	130.10514	811.2333
2005	15	21.33633	935.3667
2951	13	142.64140	956.6167

What should you do?

- Input the plausible data point?
- Delete?

Check for case sensitivity

- Code is all about case sensitivity
- Joe and joe – to us, are the same thing
- To a computer, totally different things
- Easy little trick to create comprehensive lower cases

```
data_frame$variable <- tolower(data_farm$variable)
```

```
four_sports_injury %>%  
count(knee_injury_terms)
```

```
Knee_injury_terms  n  
  
ACL, MCL, knee 6  
ACL, MCL, patella, knee 1  
ACL, knee 68  
ACL, meniscus, knee 1  
knee 4867
```

This is a lot of different cases and terms for knee injury!

How to fix this?

- Using if else logic

```
four_sports_injury <- four_sports_injury %>%  
mutate(knee_injury = ifelse(knee_injury == " ACL",  
                           'knee',  
                           knee_injury))
```

Make Sure Rows Are Uniform

- Very easy to have mix up between people or time
- Data quality 'sanity' checks are important here

```
nba_joined %>%  
filter(player_id == "aldrila01") %>%  
View()
```

	player_id	name_player	slug_team_bref	season_start_year	group_position
1	aldrila01	LaMarcus Aldridge	POR	2007	F
2	aldrila01	LaMarcus Aldridge	POR	2008	F
3	aldrila01	LaMarcus Aldridge	POR	2009	F
4	aldrila01	LaMarcus Aldridge	SAS	2017	F
5	aldrila01	LaMarcus Aldridge	SAS	2018	F
6	aldrila01	LaMarcus Aldridge	SAS	2019	F
7	aldrila01	LaMarcus Aldridge	SAS	2020	F



This was a cleaning code mix up

- Shows the importance of clear & readable code

Evaluate strings for uniformity

- Strings can be tricky!
- Extra spaces are also different cases, but to us it is no different
- Case sensitivity still applies

knee	4867
knee	7

Will use the stringr package

```
four_sports_injury$knee_injury <-  
  str_trim(four_sports_injury$knee_injury)
```

Final Thoughts

- Cleaning data is a major part of the job
- Can be difficult
- Extremely important
- Use clear and commented code to check errors